Zhou 2020 Osteosarcoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

9/11 samples (≥599 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→4.51 of 9); both lobes (Embryonic 22,156 / Cycling 10,650) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

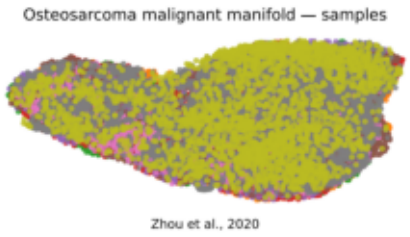
Single dominant pSom→Crest axis: pSom high over the osteoblastic body declines along pseudotime (ρ=−0.66); Crest anti-correlated (r=−0.88), rising to a terminal readout (ρ=+0.55; crest-high decile mean pt=0.93). Gast weakly structured (early programme, spent).

Gate 3 — hub mutual exclusivity → NA

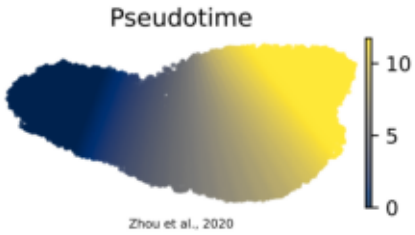
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

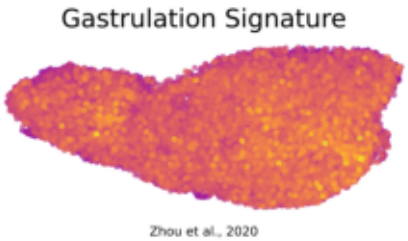
Manifold — samples



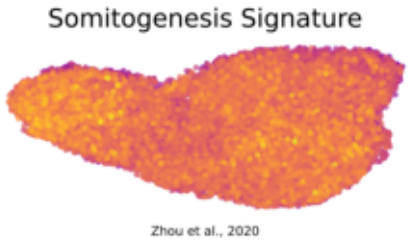
Pseudotime



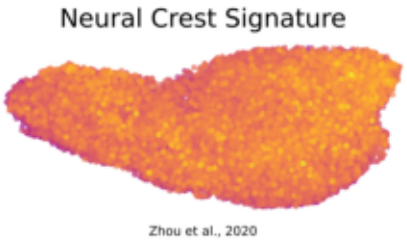
Gastrulation



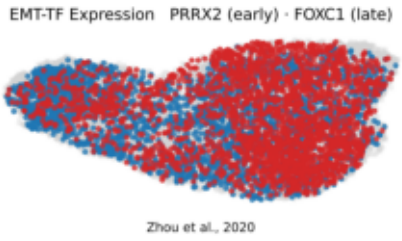
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Osteosarcoma (Zhou et al., 2020), 10x — 11 patients / 64,557 cells (master scope); malignant only: 33,459 load, 32,806 across 9 samples (≥599) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Zhou's osteosarcoma malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (osteoblastic body → chondroblastic tip), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The axis spans both proliferation lobes (orthogonal to cell cycle; module–g2m |ρ|≤0.14), so the whole manifold is kept (Bischoff/Sun no-scoping precedent). Caveat: pseudotime co-varies negatively with cycling (ρ≈−0.55) as the somitogenesis-high root lobe is the more proliferative one. EMT-TFs PRRX2 (early/root) → FOXC1 (late/terminus); canonical SNAI1/ZEB1 near the detection floor; EMT-TF end-separation weak (Δpt=0.21).

Parameter choices

Min cells/sample	≥599 (9/11; gap 1007→357)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Embryonic 22,156 / Cycling 10,650